INCH-POUND

MIL-I-85844A(AS)
14 November 1990
SUPERSEDING
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MIL-I-22352/19(AS)
10 August 1984

MILITARY SPECIFICATION

INTERCOMMUNICATION SET, AN/AIC-14
ACTIVE INTERPHONE CONTROL PANEL C-2646/AIC-14
GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance and acceptance requirements for C-2646/AIC-14 Active Interphone Control Panel, herein referred to as the C-2646.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636 Boxes, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5831

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SPECIFICATIONS (continued)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-5087	Bonding, Electrical, and Lighting Protection for Aerospace Systems
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5624	Turbine Fuel, Aviation Grades JP-4 and JP-5
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base NATO Code Number O-148
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft NATO Code Number H-537
MIL-I-85844/1	Intercommunication Set, AN/AIC-14 Amplifier Module

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking for U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of

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STANDARDS (continued)

MILITARY (continued)

MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-965	Parts Control Program
MIL-STD-45662	Calibration Systems Requirements

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

NAVAL AIR SYSTEMS COMMAND

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL 1794AS6000	Panel, Interphone Control, Active, C-2646/AIC-14

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for the associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 Item description. The C-2646 shall provide the following functions:

a. Amplification of microphone audio for outgoing interphone signals.

b. Current supply to power a microphone preamplifier or carbon microphone.

c. Keyswitch and connection of microphone generated interphone signals to the common audio output.

d. Amplification of common audio signals.

e. Mounted to an aircraft with Mounting Plate, MT-2075/AIC-14.

An outline drawing of the C-2646 is provided on Figure 1.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.4 Parts and materials.

3.4.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965, Procedure I.

3.4.2 Parts rescreening. The contractor shall provide 100 percent rescreening of active components such as but not limited to ICs, hybrids, demiconductors, and other components, as appropriate to provide a quality product. Active component parts rescreening at the electronic equipment manufacturer's plant shall consist of functional check of critical parameters at +125°C, ambient, and -54°C to the part specification.

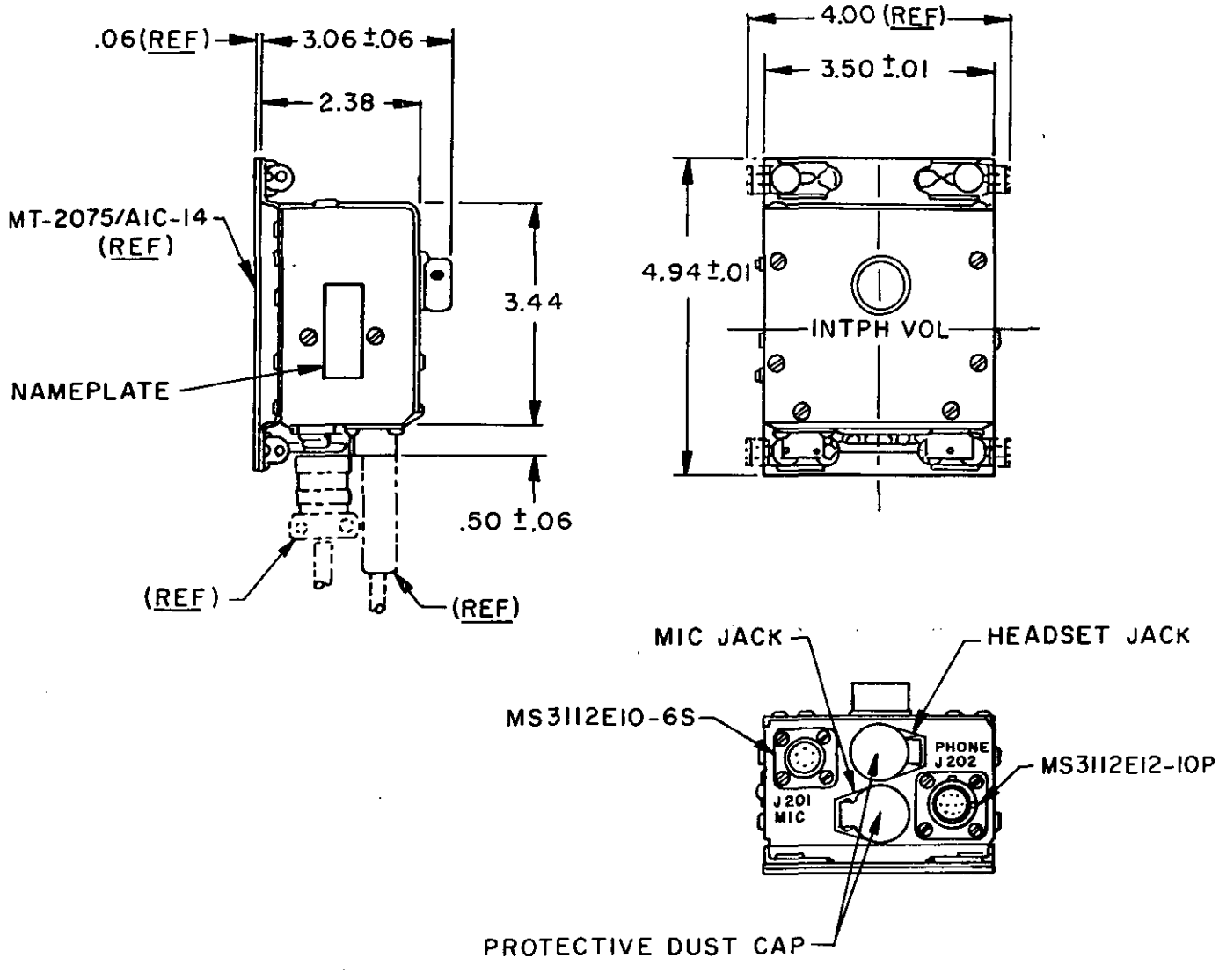
3.4.3 Supplemental requirements. In the event that the contractor proposes to alter the materials or piece parts of the C-2646 specified herein, the supplemental requirements of 3.4.3.1 through 3.4.3.3 shall be applicable for the selection of parts and materials.

3.4.3.1 Basic conformance. Parts and materials selected for use in the C-2646 shall be in accordance with MIL-E-5400, unless the parts or materials listed therein will not perform adequately in the intended applications and environments specified herein. When any parts and materials not in accordance with MIL-E-5400 must be used, refer to 3.4.1.

3.4.3.2 Toxic and corrosive fumes. The materials, as used in the manufacture of the C-2646 and under the conditions therein, shall not liberate deleterious or corrosive fumes. This shall include fungicidal agents that are used.

3.4.3.3 Recycled, virgin and reclaimed materials. Except when intended use of the C-2646 will be jeopardized by the use of reclaimed or recycled materials, these materials shall be used to the maximum extent possible.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.02 inches unless otherwise specified.

FIGURE 1. C-2646 outline.

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3.5 Design and construction. Design and construction of the C-2646 shall conform with MIL-E-5400 and DL 1794AS6000 except as otherwise specified herein. It is a condition of final acceptance that the C-2646 manufactured shall meet all the design, construction and workmanship requirements of this specification and applicable acceptance test documents.

3.5.1 Reliability.

3.5.1.1 Operational stability. The C-2646 shall operate with required performance, continuously or intermittently, for not less than 500 hours.

3.5.1.2 Operating life. The C-2646 shall have a total operating life of not less than 10,000 hours when the maintenance delineated in the maintenance plan is performed in accordance with approved procedures.

3.5.1.3 Reliability in mean-time-between-failure (MTBF). The lower test MTBF θ_1 of the C-2646 shall be not less than 500 hours.

3.5.2 Interchangeability. Interchangeability shall exist between all C-2646s and all replaceable assemblies, subassemblies and parts of the C-2646s in accordance with MIL-E-5400. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance without need for mechanical or electrical matching, alignment or adjustment of parts or circuitry.

3.5.3 Transportability. The C-2646 shall be transportable by air, rail, truck or ship when packaged as specified in Section 5.

3.5.4 Dimensions. The dimensions of the C-2646 shall be as shown on Figure 1 and as follows:

a. Height 4.94 $\pm$ 0.01 inches.

b. Width 3.50 $\pm$ 0.01 inches.

c. Depth 3.06 $\pm$ 0.06 inches.

3.5.5 Weight. The weight of the C-2646 shall be not greater than 1.5 pounds.

3.5.6 Color. The color of the C-2646 shall be as identified in DL 1794AS6000.

3.5.7 Connectors. The input and output connectors shall be as identified in DL 1794AS6000.

3.5.8 Nameplates and identification marking. Serial number assignment and nameplate approval shall be in accordance with MIL-N-18307. Identification marking shall be in accordance with MIL-STD-130. The following information shall be placed on the nameplate:

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a. Unit nomenclature (Panel, Interphone Control, Active).

b. Unit designator (C-2646/AIC-14).

c. Contract number.

d. Manufacturer's code number.

e. Equipment drawing number (30003 - 1794AS6000).

f. Technical manual identification.

g. Procuring activity (Department of the Navy - Naval Air Systems Command).

h. Serial number specified in the contract shall be stamped on the C-2646.

3.5.8.1 Serial numbers. Serial numbers shall be assigned in accordance with MIL-N-18307 for the C-2646s under contract or purchase order. The contractor shall serialize sequentially all C-2646s which are delivered under the contract.

3.5.9 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-STD-882.

3.5.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (25 ± 10°C).
b.	Altitude	Normal ground (0 - 5000 feet).
c.	Vibration	None.
d.	Humidity	Room ambient up to 90 percent relative humidity.

3.5.11 Service conditions. The C-2646 shall operate as required under any of the environmental service conditions, or any probable combination thereof, as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.5.11.1 Altitude. The C-2646 shall operate as required at any pressure up to 50,000 feet.

3.5.11.2 Fungus. The C-2646 shall withstand exposure to fungal growth. Overspraying of the C-2646 shall be prohibited.

3.5.11.3 Fluids. The C-2646 shall be unaffected by contact with the following fluids:

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a. JP-4 and JP-5 fuels (see MIL-T-5624).

b. Hydraulic fluids (see MIL-H-83282).

c. Lubricating oil (see MIL-L-7808).

d. Coolants of the fluorocarbon, silicon, silicate ester and glycol families.

3.5.12 Warm-up time. The time required for the C-2646 to warm up prior to providing voice communications shall be not greater than 10 seconds under standard conditions and 30 seconds under service conditions. The time required for the C-2646 to warm up in order to meet the performance parameters specified herein shall be not greater than 2 minutes under standard conditions and 4 minutes under service conditions.

3.5.13 Input power. The C-2646 shall meet all applicable requirements of MIL-STD-704 and shall meet specified performance requirements when energized from a power source having characteristics and limits specified in MIL-STD-704 for 28 volts dc.

3.5.13.1 Steady state power.

3.5.13.1.1 Normal steady state. The C-2646 shall meet the requirements of MIL-STD-704 for normal steady state dc input power.

3.5.13.1.2 Emergency steady state. The C-2646 shall meet the requirements of MIL-STD-704 for emergency steady state dc input power.

3.5.13.2 Transient.

3.5.13.2.1 Voltage surge. The C-2646 shall meet the requirements of MIL-STD-704 for voltage surges. The C-2646 shall not be failed for degraded performance during the voltage surges if the C-2646 returns to normal operation without damage after voltage surges pass.

3.5.13.3 Overvoltage, undervoltage and power interruption.

3.5.13.3.1 Overvoltage. The C-2646 shall meet the requirements of MIL-STD-704 for overvoltage. The C-2646 shall not be failed for degraded performance during overvoltage exposure if the C-2646 returns to normal operation without damage after the exposure.

3.5.13.3.2 Undervoltage. The C-2646 shall meet the requirements of MIL-STD-704 for undervoltage. The C-2646 shall not be failed for degraded performance during undervoltage exposure if the C-2646 returns to normal operation without damage after the exposure.

3.5.13.3.3 Power interruption. The C-2646 shall meet the requirements of MIL-STD-704 for power interruption. The C-2646 shall not be failed for degraded performance during power interruption if the C-2646 returns to normal operation without damage after the exposure.

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3.5.14 Electromagnetic Interference (EMI). The C-2646 shall comply with the requirements of MIL-STD-461, Class A1b equipment, as modified herein. These requirements shall be met in all operating modes and at all operating frequencies of the C-2646 at standard conditions specified in 3.5.10. EMI testing shall be accomplished in accordance with the procedures of MIL-STD-462 to demonstrate compliance with requirements of MIL-STD-461. EMI design guidelines shall be in accordance with AD-1115.

3.5.14.1 Conducted and radiated emissions.

3.5.14.1.1 Conducted emission, power and interconnecting leads. Conducted emissions from the C-2646 power and interconnecting leads shall be within the limits of MIL-STD-461, Requirement CE03.

3.5.14.1.2 Conducted emissions, power leads, spikes, time domain. Conducted emissions from the C-2646 power leads shall be within the limits of MIL-STD-461, Requirement CE07.

3.5.14.1.3 Radiated emissions electric field 14 kHz to 10 GHz. Radiated emissions from the C-2646, electric field, between 14 kHz and 10 GHz, shall be within the limits of MIL-STD-461, Requirement RE02. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized waves.

3.5.14.2 Conducted and radiated susceptibility.

3.5.14.2.1 Conducted susceptibility, power leads, 30 Hz to 50 kHz. The C-2646 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to electromagnetic energy injected onto its power leads less than or equal to the values of MIL-STD-461, Requirement CS01.

3.5.14.2.2 Conducted susceptibility, power leads, 50 kHz to 400 MHz. The C-2646 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to 1 volt from a 50-ohm source across a calibrating resistor as specified in MIL-STD-461, Requirement CS02. This requirement shall apply to all neutrals and grounds whether or not they are internally returned to chassis ground.

3.5.14.2.3 Conducted susceptibility, spikes, power leads. The C-2646 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when spikes each having the waveform specified in MIL-STD-461, Requirement CS06, are applied to the ac and dc power input leads.

3.5.14.2.4 Radiated susceptibility, magnetic induction field, spikes and power frequencies. The C-2646 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to radiated signals specified in MIL-STD-461, Requirement RS02.

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3.5.14.2.5 Radiated susceptibility, electric field, 14 kHz to 40 GHz. The C-2646 shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to the radiated electric fields specified in MIL-STD-461, Requirement RS03, except field levels and modulation shall be as specified in Table I.

3.5.14.3 Susceptibility interpretation. The criteria of 3.5.14.3.1 and 3.5.14.3.2 shall be applied to determine whether the unit is susceptible.

3.5.14.3.1 Susceptibility to stray magnetic fields. The C-2646 shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 Hz to 2800 Hz, the properly terminated output from the amplifier circuit shall be not greater than 10.0 millivolts at 400 Hz and shall be not greater than a linear increase of 16μV/Hz up to 2800 Hz.

3.5.14.3.2 Susceptibility. The C-2646 shall be designed to resist audio interference including power supply pickup. The C-2646 shall exhibit audio outputs not greater than -70 dB within the audio pass band of 100 Hz to 10 kHz when subjected to the radiated and conducted signal exposures required herein.

3.5.14.4 Ground plane interference. The C-2646 shall be capable of specified performance when subjected to aircraft chassis noise between the interfacing WRAs.

3.5.14.5 Grounding requirements. The grounding system between WRAs of the equipment shall be compatible and consistent with the following aircraft grounding scheme. All WRAs shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, minimize EMI, and to protect personnel from electrical hazards.

3.5.14.6 Bonding. A means of electrically bonding the equipment to the aircraft chassis shall be provided and shall comply with the requirements of MIL-B-5087. Bonding resistance between the bonding surface and all parts of the equipment case shall be not greater than 2.5 milliohms.

3.5.15 Interface.

3.5.15.1 Signal interface. The C-2646 input and output signals shall have the characteristics specified in Tables II, III and IV.

3.5.15.2 Operator interface.

3.5.15.2.1 Controls. The C-2646 shall be provided with an Interphone Volume Control (INTPH VOL). The INTPH VOL control shall be a rotary-type control, providing an adjustment range of 33 + 3 dB. The amplitude of incoming radio receiver signals shall not be affected by more than ±1 dB when the INTPH VOL control is adjusted throughout its range.

3.6 Performance.

3.6.1 Input characteristics. The input signals listed in Table II shall be provided having the characteristics specified in 3.6.1.1 through 3.6.1.5.

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TABLE I. RS03 signal (continuous application test levels).

Frequency (MHz)	Modulation				Power Density (mW/cm^2) Peak 5/	Power Density (mW/cm^2) Ave	v/M(Ref) 6/
0.014-850	AM 90% 1 kHz Sine Wave				--	0.66	50
215-225	PAM 1/	20 usec 2/	PW 3/	200 pps 4/	0.126	--	22
215-225	PAM	200 usec	PW	200 pps	0.126	--	22
400-450	PAM	4 usec	PW	300 pps	0.200	--	28
850-940	PAM	10 usec	PW	1000 pps	0.316	--	35
850-940	PAM	125 usec	PW	300 pps	0.316	--	35
940-1215	PAM	10 usec	PW	1000 pps	0.106	--	20
1215-1365	PAM	10 usec	PW	3000 pps	0.106	--	20
1365-2900	PAM	1 usec	PW	1000 pps	0.160	--	25
2900-3100	PAM	3 usec	PW	1000 pps	1.60	--	78
2900-3100	PAM	70 usec	PW	200 pps	1.60	--	78
3100-3500	PAM	10 usec	PW	3000 pps	3.18	--	110
3100-3500	PAM	50 usec	PW	3000 pps	3.18	--	110
3500-9000	PAM	1 usec	PW	1000 pps	0.636	--	50
9000-14000	PAM	0.2 usec	PW	1000 pps	0.106	--	20
14000-18000	PAM	0.2 usec	PW	50000 pps	0.106	--	20
33200	PAM	0.2 usec	PW	2000 pps	0.106	--	20

NOTES:

1/ Pulse amplitude modulated.
2/ Microseconds.
3/ Pulse width.
4/ Pulses per second.
5/ Peak power density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; for example, average power density within the pulse envelope.
6/ Volts per meter equivalent is shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

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TABLE II. Input signals.

Signal Name	Characteristics
Microphone Audio (MIC AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 1.0 Vac.
Spare Input	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 1.0 Vac.
Common Audio (COM AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 27 Vac.
Keyline	Relay logic signal: (1) Open and (2) short to ground.
Power (PWR)	+27.5 ± 0.5 Vdc.

TABLE III. Output signals.

Signal Name	Characteristics
Keyline	Relay logic: (1) Open and (2) short to ground.
Headset (HDST)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 27 Vac.
Common Audio (COM AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 27 Vac.

TABLE IV. Grounds.

Signal Name	Characteristics
Power Return (PWR GND)	Primary power is specified as the power delivered to the equipment by the aircraft generating and distribution system.
System Return (SYS RTN)	The system ground shall not be electrically connected to power ground.
Microphone Return (MIC RTN)	Signal return to system ground.
Headset Return (HDST RTN)	Signal return to system ground.

3.6.1.1 Microphone audio (MIC AUDIO).

3.6.1.1.1 Signal frequency (MIC AUDIO). The MIC AUDIO signal input shall accept drive signals which range from 300 to 7000 Hz.

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3.6.1.1.2 Signal amplitude (MIC AUDIO). The MIC AUDIO signal inputs shall accept drive signals which range in amplitude from 0.01 to 1 Vac. The circuitry shall be capable of withstanding overvoltage inputs of up to 2 Vac without sustaining damage.

3.6.1.1.3 Input impedance (MIC AUDIO). The impedance of the MIC AUDIO signal input shall be 100 ohms ± 10 percent.

3.6.1.1.4 Microphone current (MIC AUDIO). The MIC AUDIO input line shall be capable of supplying 60 + 10 milliamperes to the microphone circuit for carbon microphones or as a source of power to equivalent dynamic microphone and associated preamplifier combinations.

3.6.1.2 Spare input. The Spare input shall be applied to the audio amplifier (see MIL-I-85844/1) through the INTPH VOL control and the key relay (when deenergized). The Spare input shall have an input impedance of 480 ohms ± 10 percent with the INTPH VOL control fully clockwise and keyline open.

3.6.1.3 Common audio (COM AUDIO). The COM AUDIO input shall be used as an input or output. The COM AUDIO input is summed with the Spare input, passed through the INTPH VOL control, and applied to the amplifier module when the key relay is deenergized. See 3.6.2.2 for output characteristics. The COM AUDIO input shall have the characteristics specified in 3.6.1.3.1 through 3.6.1.3.3.

3.6.1.3.1 Signal frequency (COM AUDIO). The COM AUDIO input shall accept drive signals which range from 300 to 7000 Hz.

3.6.1.3.2 Signal amplitude (COM AUDIO). The COM AUDIO input shall accept input drive signals which range in amplitude from 0 to 27 Vac.

3.6.1.3.3 Input Impedance (COM AUDIO). The COM AUDIO input shall be 62,000 ohms ± 10 percent.

3.6.1.4 Keyline. The Keyline signal is a relay logic drive signal having two states based on the position of the operator's keyswitch:

a. Open.

b. Short to ground.

The signal shall be applied to the input circuitry to activate the keyswitch relay function. Signal current shall be not greater than 80 ma.

3.6.1.5 Power (PWR). The C-2646 shall accept 27.5 ± 0.5 Vdc power, while the input current shall be not greater than 300 ma under any condition.

3.6.2 Output characteristics. The output signals listed in Table III shall be provided having the characteristics specified in 3.6.2.1 through 3.6.2.3.6.

3.6.2.1 Keyline. The Keyline output shall be a relay logic signal having two states:

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a. Open.

b. Short to ground.

The Keyline signal shall be a short to ground when transmitter keying is desired as evidenced by actuation of the microphone keyswitch.

3.6.2.2 Common audio (COM AUDIO). The COM AUDIO line shall provide amplified MIC AUDIO signals when the key relay is energized by a logic LO on the Keyline input. The COMMON AUDIO output shall have the characteristics specified in 3.6.2.2.1 through 3.6.2.2.7. The COM AUDIO output may also be used as an input (see 3.6.1.3).

3.6.2.2.1 Signal frequency (COM AUDIO). The COM AUDIO output amplitude shall be in accordance with the response curves shown on Figure 2.

3.6.2.2.2 Signal amplitude/gain (COM AUDIO). The COMMON AUDIO output shall be greater than a nominal input of 0.06 V at 1000 Hz applied to MIC AUDIO input by 35 + 2 dB (2.68 - 4.25 V). The input/output characteristic (gain) of the amplifier shall be linear within ±1 dB when the input level is varied between 0.01 and 0.10 volts rms.

3.6.2.2.3 Distortion (COM AUDIO). The total harmonic distortion over the frequency range of 300 to 7000 Hz and microphone input amplitude range of 0.01 to 0.10 V shall be not greater than 5 percent under standard conditions. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 8 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

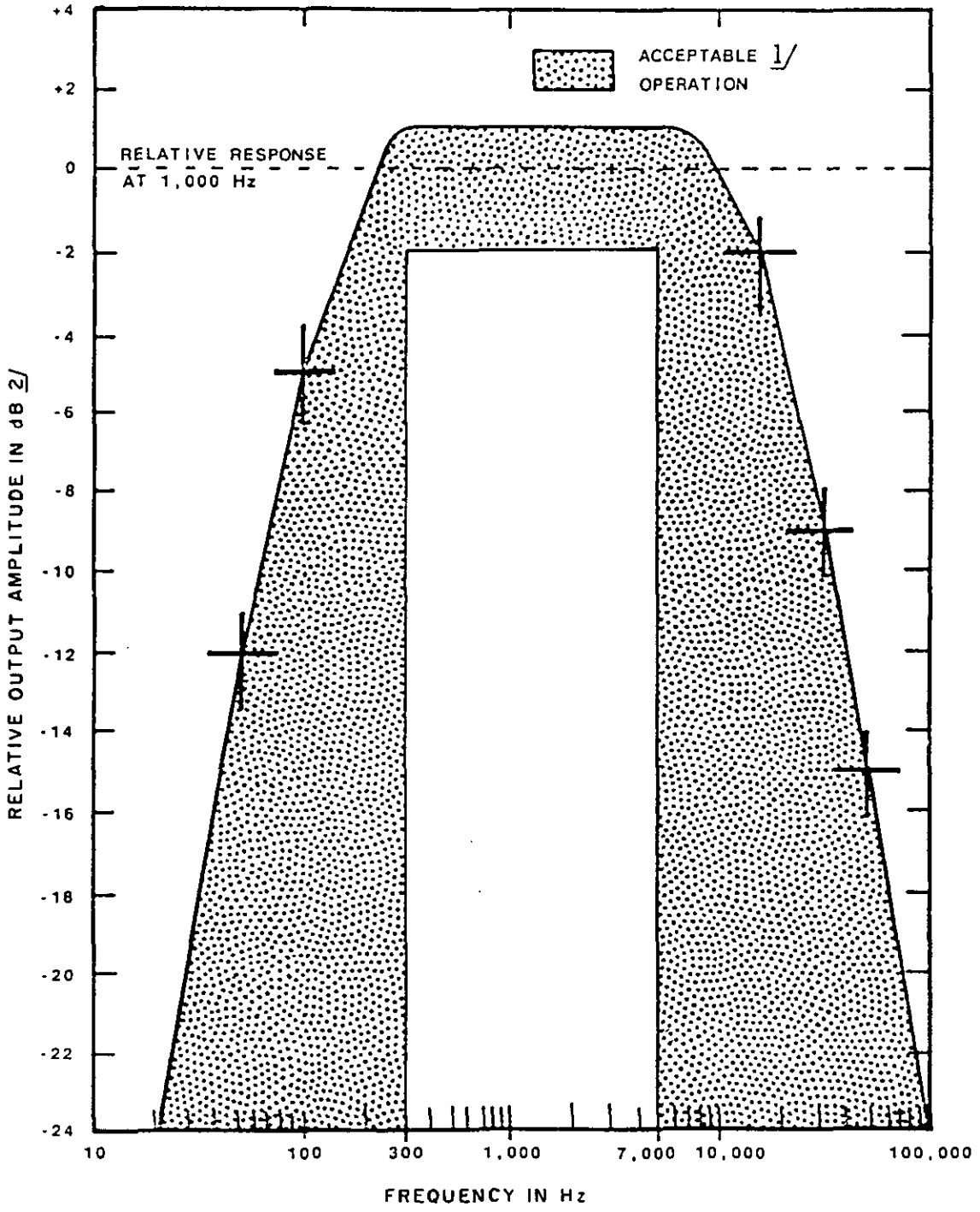
3.6.2.2.4 Output power (COM AUDIO). The output of the amplifier shall be capable of delivering 1 watt into a 600 ohm resistive load.

3.6.2.2.5 Output stability (COM AUDIO). The output shall show no evidence of regeneration or oscillation throughout the environmental envelope of MIL-E-5400 for Class 2 equipment under any combination of input-output open, short or complex impedance terminations. The C-2646 shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specified limits when properly terminated.

3.6.2.2.6 Internal noise (COM AUDIO). The amplifier shall have an output of less than 5 millivolts when operated with the input shorted to signal ground throughout the environmental envelope of MIL-E-5400 for Class 2 equipment.

3.6.2.2.7 Susceptibility to stray magnetic fields (COM AUDIO). The amplifier module shall be shield or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2800 Hz, the output from a properly terminated amplifier circuit shall be not greater than 10.0 millivolts at 400 Hz. Between 400 and 2800 Hz, the output shall be not greater than a linear increase up to 50 millivolts at 2800 Hz.

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1/ Termination into 600 ohms.
2/ Output amplitude is measured with respect to the amplitude at 1000 Hz.

FIGURE 2. Output response curves.

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3.6.2.3 Headset (HDST).

3.6.2.3.1 Signal frequency (HDST). The HDST output amplitude shall be in accordance with the response curves shown on Figure 2.

3.6.2.3.2 Signal amplitude/gain (HDST). The HDST output shall be greater than a nominal input of 0.06 V at 1000 Hz by 1 + 2 dB (0.053 - 0.085 V) when terminated in a 600 ohm resistive load, the key relay is deenergized, INT Vol control is fully clockwise, and COM AUDIO is used as an input (see 3.6.1.3).

3.6.2.3.3 Output power (HDST). The output of the amplifier shall be capable of delivering 1 watt into a 600 ohm resistive load.

3.6.2.3.4 Distortion (HDST). The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 5 percent under standard conditions. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.6.2.3.5 Output stability (HDST). The output shall show no evidence of regeneration or oscillation throughout the environmental envelope of MIL-E-5400 for Class 2 equipment under any combination of input-output, open, short or complex impedance terminations. The C-2646 shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specified limits when properly terminated.

3.6.2.3.6 Susceptibility to stray magnetic fields. The amplifier module shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2800 Hz, the properly terminated output from the amplifier circuit shall be not greater than 10.0 millivolts at 400 Hz. Between 400 and 2800 Hz, the output shall be not greater than a linear increase up to 50 millivolts at 2800 Hz.

3.7 SRAs. The C-2646 shall contain an audio amplifier SRA having a gain of 43 + 2 dB at 1000 Hz. Performance of the amplifier shall be in accordance with MIL-I-85844/1.

3.8 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9, and the following:

a. Printed wiring board coatings shall be free of dirt and metal particles.

b. Chemical coated and painted surfaces shall have uniformly coated surfaces.

c. Metal parts shall be free of burrs or sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment and inspection facility. The contractor shall ensure that inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of the required inspections. Inspection equipment used in the inspections specified herein shall be maintained in calibration by a facility acceptable to the procuring activity which meets the requirements of MIL-STD-45662.

4.1.3 Quality assurance terms. Quality assurance terms used in this specification shall be as specified in MIL-STD-109.

4.1.4 Classification of inspections. The inspection requirements specified herein are classified as:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

c. Special inspections (see 4.4).

4.1.5 Inspection conditions. Unless otherwise specified in the contract (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in 3.3.10.

4.2 First article inspection. First article inspections specified in 3.3 shall be conducted by the contractor on not less than one C-2646 representative of the processes, methods and techniques used on the C-2646s under contract. The first article inspection shall include all inspections required to verify compliance with all electrical and environmental requirements of this specification. The Government quality assurance representative and the procuring activity shall be notified when inspections are to be conducted to allow monitoring if desired. Contractors not having adequate

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facilities to conduct all the required inspections may obtain the services of a commercial inspection laboratory only when said laboratory is acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that the C-2646s conform to specified requirements. First article inspection shall include the following tests:

a. Altitude (see 4.2.1.1).

b. High temperature (see 4.2.1.2).

c. Low temperature (see 4.2.1.3).

d. Humidity (see 4.2.1.4).

e. Salt fog (see 4.2.1.5).

f. Vibration (see 4.2.1.6).

g. Mechanical shock (see 4.2.1.7).

h. Electromagnetic interference (see 4.2.2).

i. Electrical requirements inspection (see 4.2.3).

4.2.1 Environmental tests. Environmental tests shall be in accordance with MIL-STD-810, except as modified herein. The C-2646 shall operate as required herein, during (unless as otherwise indicated) and after each environmental test. The C-2646 shall not sustain either physical damage or performance degradation during or as a result of exposure to the tests. The C-2646 shall be allowed a +4 dB gain (see 3.6.2.2.2 and 3.6.2.3.2) when subjected to hot/cold temperature cycling. When the unit is returned to standard conditions, the gain shall remain ±2 dB for HDST and ±2 dB for COM AUDIO.

4.2.1.1 Altitude. The C-2646 shall be subjected to an altitude test in accordance with MIL-STD-810, Method 500.2, Procedure II. The test altitude shall be limited to 50,000 feet and the rate of change shall be not greater than 2000 feet per minute.

4.2.1.2 High temperature. The C-2646 shall be subjected to high temperature operating test in accordance with MIL-STD-810, Method 501.2, Procedure II, except steps 3 through 6 of the procedure shall be deleted. The chamber temperature for step 7 shall be +71°C.

4.2.1.3 Low temperature. The C-2646 shall be subjected to low temperature operating tests in accordance with MIL-STD-810, Method 502.2, Procedure II. The chamber temperature shall be -54°C.

4.2.1.4 Humidity. The C-2646 shall be subjected to humidity test in accordance with MIL-STD-810, Method 507.2, Procedure III, Figure 507.2-3. The test duration shall be 10 cycles.

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4.2.1.5 Salt fog. The C-2646 shall be subjected to salt fog test in accordance with MIL-STD-810, Method 509.2, Procedure I. The test duration shall be 48 hours. Following the 48-hour drying period, the C-2646 shall meet performance requirements specified herein.

4.2.1.6 Vibration. The C-2646 shall be subjected to random vibration exposure in accordance with MIL-STD-810, Method 514.3, Procedure I for Category 5 equipment. Both functional and endurance tests shall be conducted using the vibration spectra shown on Figure 3. Vibration shall be applied sequentially along each of three mutually perpendicular axes for one-half hour at the functional level, and one hour at the endurance level followed by one-half hour at the functional level. The C-2646 shall operate as required during exposure to the functional level and immediately before and after exposure to the endurance level.

4.2.1.7 Shock. The C-2646 shall be subjected to shock tests in accordance with MIL-STD-810.

4.2.1.7.1 Functional. The C-2646 shall withstand three shocks in each direction along three mutually perpendicular axes in accordance with MIL-STD-810, Method 516.3, Procedure I, and Figure 516.3-4 for operational test.

4.2.1.7.2 Crash hazard. The C-2646 shall be subjected to two shocks in each direction along three mutually perpendicular axes in accordance with MIL-STD-810, Method 516.3, Procedure V and Figure 516.3-4 for crash hazard test. There shall be no failure of the mounting attachment and the C-2646 shall remain in place and not create a hazard. Bending and distortion shall be permitted.

4.2.1.7.3 Bench handling. The C-2646 shall be tested in accordance with MIL-STD-810, Method 516.3, Procedure VI.

4.2.2 Electromagnetic interference (EMI). The C-2646 shall be subjected to the EMI tests specified in 4.2.2.1 through 4.2.2.6 in accordance with the test methods of MIL-STD-462 to verify that C-2646 meets the requirements of MIL-STD-461, Class A1, Category A1b, for the EMI tests herein.

4.2.2.1 Conducted and radiated emission.

4.2.2.1.1 Conducted emissions, power and interconnecting leads, 0.15 to 50 MHz. Conducted emissions measurements shall be made over the frequency range of 0.15 to 50 MHz on power leads in accordance with MIL-STD-462, Test Method CE03, and interconnecting leads in accordance with MIL-STD-462, Test Method CE04.

4.2.2.1.2 Conducted emissions power leads, spikes, time domain. The contractor shall devise a test to demonstrate compliance with the requirements of 3.5.14.1.2.

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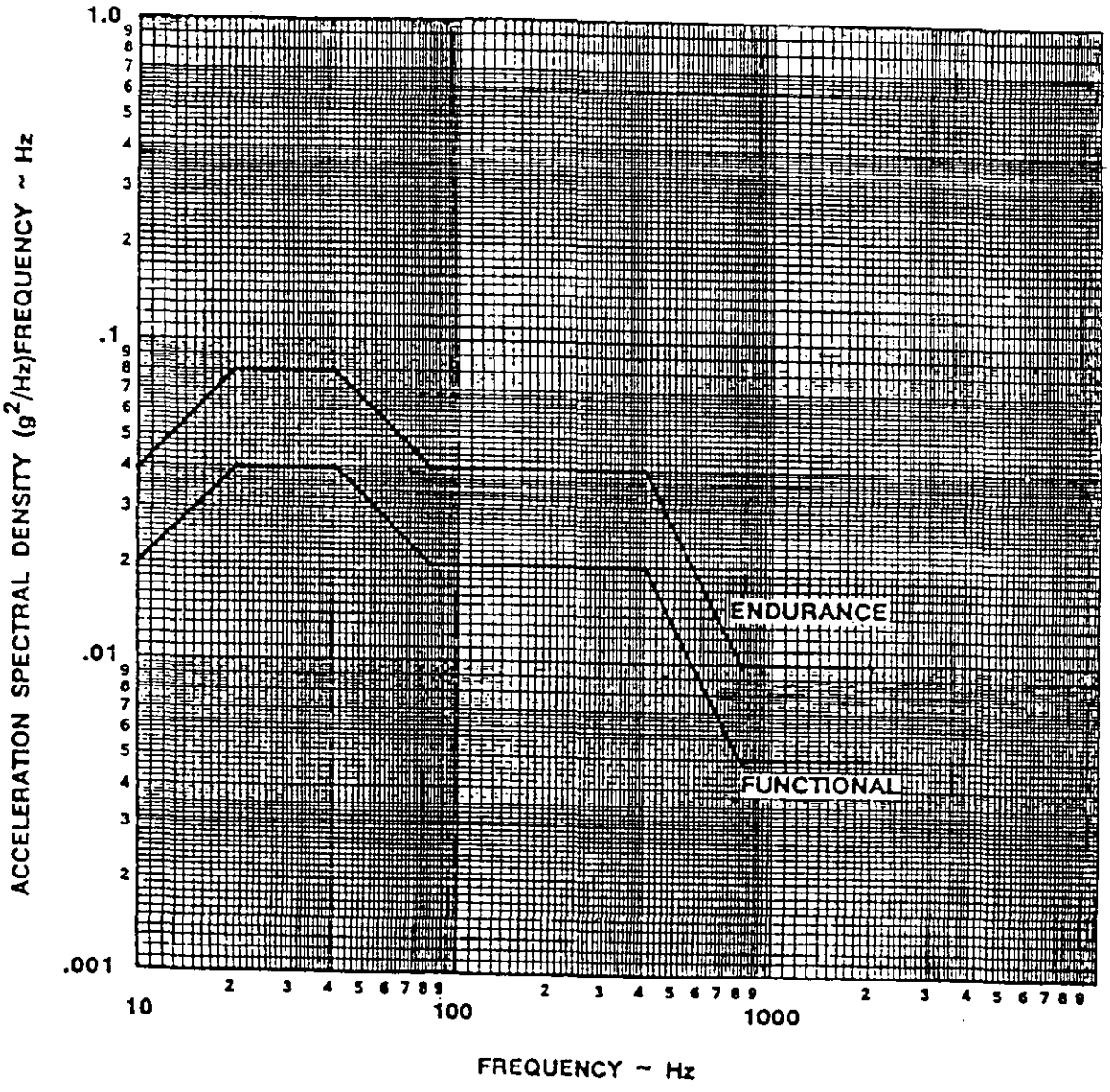


FIGURE 3. Vibration test spectrum.

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4.2.2.1.3 Radiated emissions, magnetic field, 0.03 to 50 kHz. Radiated emissions emanating from the C-2646 shall be measured over frequency range of 0.03 to 50 kHz (magnetic field) in accordance with MIL-STD-462, Test Method RE02.

4.2.2.2 Conducted and radiated susceptibility.

4.2.2.2.1 Conducted susceptibility, power leads, 30 Hz to 50 kHz. The C-2646 shall be subjected to conducted susceptibility tests by having signals over the range of 30 Hz to 50 kHz injected on the power leads in accordance with MIL-STD-462, Test Method CS01. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.2 Conducted susceptibility, power leads, 50 kHz to 400 MHz. The C-2646 shall be subjected to conducted susceptibility tests by having signals over the range of 50 kHz to 400 MHz injected on the power leads in accordance with MIL-STD-462, Test Method CS02. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.3 Conducted susceptibility, spikes, power leads. The C-2646 shall be subjected to conducted susceptibility tests consisting of superimposing a 32 volt maximum spike on the power line voltage waveform in accordance with MIL-STD-462, Test Method CS06. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.4 Radiated susceptibility magnetic induction field spikes and power frequencies. The C-2646 shall be subjected to radiated susceptibility tests consisting of spikes and power frequencies in accordance with MIL-STD-462, Test Method RS02. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.1.

4.2.2.2.5 Radiated susceptibility, electric field, 14 kHz to 40 GHz. The C-2646 shall be subjected to radiated susceptibility tests over the frequency range specified in Table I, electric field, in accordance with MIL-STD-462, Test Method RS03.

4.2.2.2.6 Ground plane interference. The contractor shall devise a test to demonstrate compliance with the requirements of 3.5.14.4. This test shall be included as a part of the EMI test plan.

4.2.3 Electrical requirements inspection. The contractor shall conduct inspections necessary to demonstrate compliance to the electrical requirements of Section 3 and DL 1794AS6000. Demonstrations are to include but not be limited to the tests herein.

4.2.3.1 Warm-up time. Time required for the C-2646 to stabilize after being energized at standard and environmental conditions shall be monitored to assure that the C-2646 meets the requirements of 3.5.12.

4.2.3.2 Input power.

4.2.3.2.1 Steady state power.

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4.2.3.2.1.1 Normal steady state. The C-2646 shall be checked operationally with input voltage set at 22 volts dc and 29 volts dc in accordance with MIL-STD-704 for normal steady state test. The C-2646 shall be operated long enough to assure that the C-2646 operates as required with each of these input voltages applied.

4.2.3.2.1.2 Emergency steady state. The C-2646 shall be checked operationally with input voltage set at 16 volts dc and 29 volts dc in accordance with MIL-STD-704 for emergency steady state test. The C-2646 shall be operated long enough to assure that the C-2646 operates as required with each of these input voltages applied.

4.2.3.2.2 Transient testing.

4.2.3.2.2.1 Voltage surge. The C-2646 shall be subjected to input voltage surges of 18 to 50 volts dc for 0.013 to 0.042 second (transient duration) in accordance with MIL-STD-704 for transient test. The C-2646 operation shall be observed for permanent degradation.

4.2.3.2.3 Overvoltage, undervoltage and power interruption.

4.2.3.2.3.1 Overvoltage. The C-2646 shall be subjected to input voltage of 50 volts dc for 0.045 second in accordance with MIL-STD-704 for overvoltage test. The C-2646 operation shall be observed for permanent degradation.

4.2.3.2.3.2 Undervoltage. The C-2646 shall be subjected to a change of input voltage from normal to 0 volt dc for 5 seconds in accordance with MIL-STD-704 for undervoltage test. The C-2646 operation shall be observed for permanent degradation.

4.2.3.2.3.3 Power interruption. The C-2646 shall be subjected to input power interruption of 50 ms in accordance with MIL-STD-704 for power interruption test. The C-2646 operation shall be observed for permanent degradation.

4.3 Quality conformance inspection.

4.3.1 Individual inspections. Each production C-2646 submitted for acceptance shall be subjected to the required inspections specified herein. These inspections shall be adequate to determine compliance with the requirements for material, workmanship, operational performance and environmental stress screening (ESS).

a. Examination of product (see 4.3.1.1).

b. Operational inspection (see 4.3.1.2).

c. Environmental stress screening (ESS) (see 4.3.1.3).

d. Production reliability acceptance test (PRAT) (see 4.3.1.4).

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4.3.1.1 Examination of product. Each production C-2646 shall be examined carefully to ensure compliance with requirements of workmanship, marking, interchangeability, design and construction.

4.3.1.2 Operational inspections. Each production C-2646 shall be subjected to an operational inspection. The inspection shall demonstrate and test all modes of operation; all input/output signals shall be tested and the results recorded. During the inspection, sufficient characteristics shall be measured and recorded to assure that the C-2646s submitted for acceptance meet all the requirements of this specification.

4.3.1.2.1 Input impedance. Verify that the input impedances are in accordance with 3.6.1.1.3, 3.6.1.2 and 3.6.1.3.3 herein. Impedances shall be measured at 1000 Hz. The input impedance shall be measured using the resistance substitution method.

4.3.1.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.6.2.2.2 and 3.6.2.3.2.

4.3.1.2.3 Frequency response. Holding the input signal amplitude constant, vary the frequency between 300 - 7000 Hz. Verify that the amplifier gain meets the requirements of 3.6.2.2.1 and 3.6.2.3.1 herein.

4.3.1.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and 0.1 Vac, verify that the amplifier meets the requirements of 3.6.2.2.2.

4.3.1.2.5 Output power. With the line out terminated with 600 ohms, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.6.2.3.5 aNd 3.6.2.2.4 herein.

4.3.1.2.6 Distortion. Verify the requirements of 3.6.2.2.3 and 3.6.2.3.4.

4.3.1.2.7 Stability. Verify the requirements of 3.6.2.2.5 and 3.6.2.3.6.

4.3.1.2.8 Internal noise. Verify the requirements of 3.6.2.2.6.

4.3.1.2.9 Susceptibility to stray magnetic fields. Verify the performance of 3.6.2.2.7 and 3.6.2.3.7 herein.

4.3.1.2.10 Controls. Verify control operation of 3.5.15.2 herein.

4.3.1.3 Environmental stress screening (burn-in). An environmental stress screening inspection shall be required on all production C-2646s. Prior to and after each submission to the screening inspection, a full electrical performance inspection under standard conditions shall be performed on each production C-2646. Each production C-2646 shall be subjected to a workmanship random vibration of 6.3 g's rms with a frequency of 20 to 2000 Hz, as measured on the vibration fixture at the mounting points (see Figure 4). The vibration duration shall be not less than 10 minutes (failure free) in the axis that is perpendicular to the majority of the printed wiring boards. The production C-2646 shall be mounted to the vibration fixture by its normal

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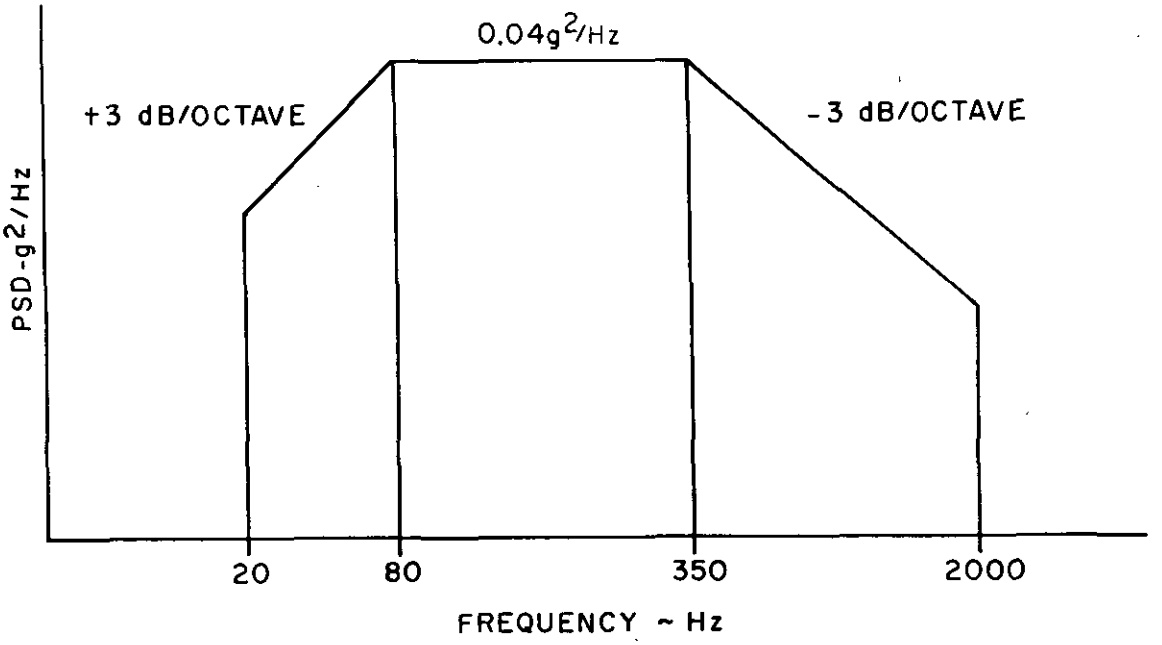


FIGURE 4. Workmanship random vibration.

mounting means so as to meet the vibration requirements specified herein. The production C-2646 shall be continuously energized and electrically monitored for the duration of the inspection. After the vibration inspection each production C-2646 shall be subjected to a minimum of 72 hours of thermal stress screening of which not less than 45 hours shall be with the C-2646 energized. The thermal stress screening inspection shall consist of nine 8-hour temperature cycles of which each cycle contains a 3-hour cold period at -54°C (C-2646 deenergized) and a 5-hour hot period at +71°C (C-2646 energized). Each production C-2646 shall be deenergized for the duration of the cold period except for the time needed to perform the electrical performance inspection specified herein (see Figure 5). Each production C-2646 shall be energized for the full duration of the hot period. An electrical performance inspection shall be performed once daily during both hot and cold periods, after the production C-2646s have stabilized at inspection temperatures. The last two temperature cycles shall be failure free. The thermal stress screening inspection chamber's temperature change rate shall be greater than 10°C per minute. The contractor shall be afforded the option of performing the vibration inspection concurrently with the temperature cycling portion of the screening inspection or as a stand-alone inspection.

4.3.1.4 Production reliability acceptance test (PRAT). Following the environmental stress screening inspection of 4.3.1.3 each C-2646 shall be subjected to a production reliability acceptance test as a means for verifying the MTBF requirement of 3.5.1.3 (500 hours). Only those C-2646s that have successfully completed both the environmental stress screening inspection and the production reliability acceptance test shall be delivered to the procuring

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activity. The production reliability acceptance test shall be in accordance with MIL-STD-781, Test Plan XVIIIC. Each C-2646 shall accumulate not less than 45 operating hours (last 3 cycles failure free) under environmental cycling conditions as shown on Figure 5. No particular C-2646 shall be allowed to accumulate more than 90 hours of production reliability acceptance test time.

4.3.1.5 Determination of compliance. The total production reliability acceptance test time and chargeable failures will be plotted on the Continue Test/Reject Criteria Graph of MIL-STD-781, Test Plan XVIIIC, using a lower test MTBF O_1 of 500 hours. The upper test MTBF O_o shall be 1000 hours (see Figure 6). If the plot of test failures versus total test time enters the reject region, testing shall be suspended and the procuring activity along with the quality assurance representative shall be notified.

4.4 Special inspections. Procuring activity reserves the right to perform any one or group of inspections set forth in this specification when such inspections are deemed necessary by the procuring activity to assure production item conformance to the specified requirements. Conditions that warrant special inspections shall include:

a. First production C-2646 incorporating a design or material change.

b. Whenever production is interrupted for more than 1 year.

c. Whenever failure reports or other information indicate that additional inspections are required.

d. Whenever it is deemed necessary to assure production C-2646s are equal to the previously approved first article inspection C-2646 in design, construction, workmanship, performance and quality.

e. As specified in the contract or purchase order (see 6.2.1).

4.5 Classification of failures. Should a failure occur during any inspection, the following action shall be taken:

a. Determine the cause of failure.

b. Determine whether the failure is an isolated case or design or production process defect.

c. Implement corrective action intended to reduce the possibility of the same failure(s) occurring in future inspections.

d. Where practical, include additional sequences during the quality conformance acceptance inspections until assurance is obtained that the defect has been corrected.

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PRODUCTION RELIABILITY ACCEPTANCE TEST
ENVIRONMENTAL TEMPERATURE CYCLE
C-2646/AIC-14

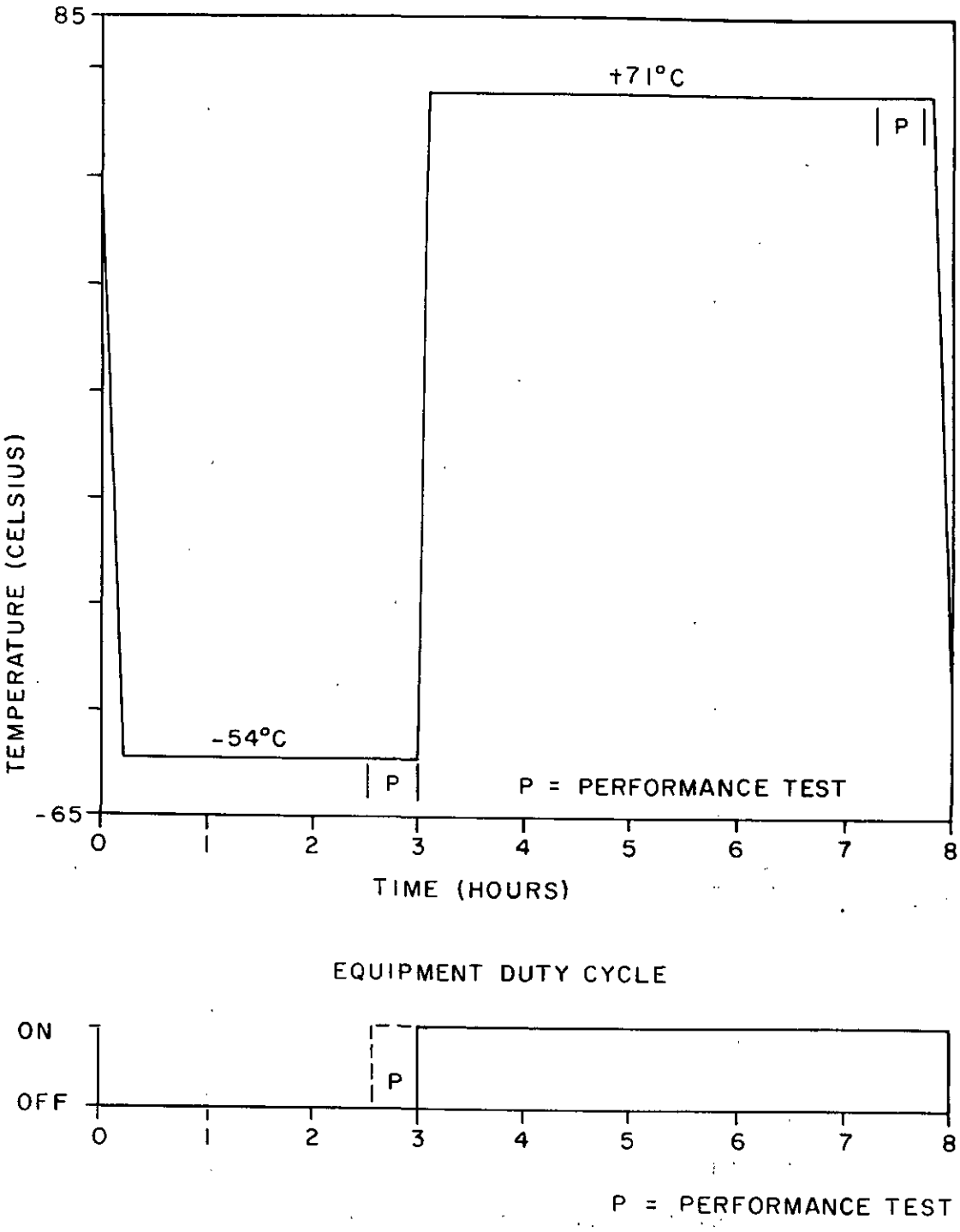


FIGURE 5. Environmental cycling conditions.

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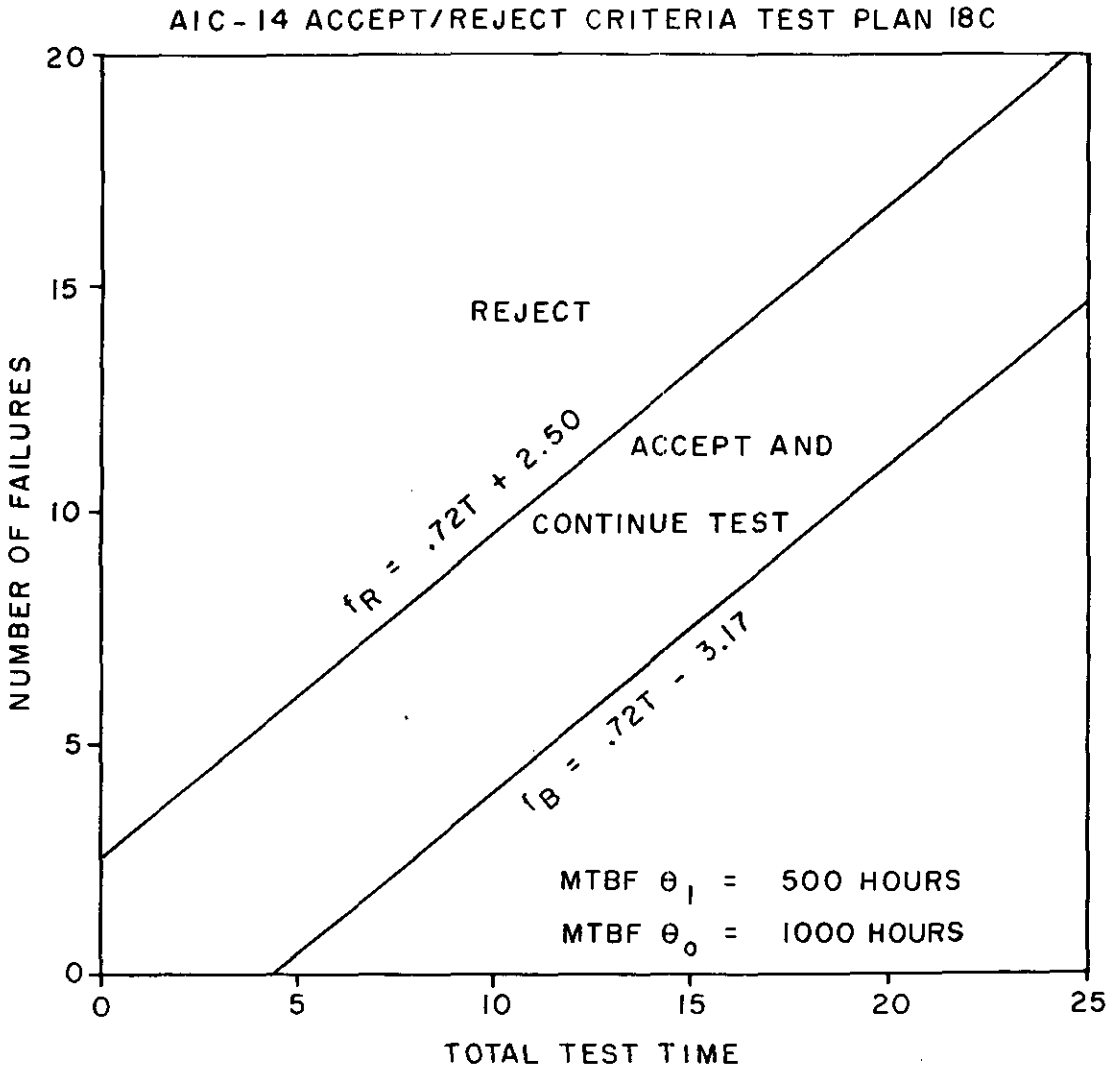


FIGURE 6. Continue test/reject criteria graph.

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4.6 Methods of examination and inspection. All inspections shall be conducted in accordance with Government approved procedures. When approved inspection procedures are available from previous contracts, they may be provided and used upon authorization of the procuring activity. The Government reserves the right to modify procedures or add inspections when it is deemed necessary in order to assure C-2646s are in compliance with this specification and the contract.

5. PACKAGING

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as follows.

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order (see 6.2.1), preservation-packaging shall be in accordance with MIL-P-116, Method III.

5.3 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), packing shall be level A. The packaged equipments shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-636.

5.3.1 Unitized loads. Unitized loads, commensurate with the level of packing specified herein or in the contract (see 6.2.1), shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantity to the greatest extent possible.

5.4 Marking. In addition to any special marking required by the contract (see 6.2.1) each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The C-2646 Active Interphone Control Panel is intended to be a major component of the AN/AIC-14 Intercommunications Set. The C-2646 is to provide amplification of audio signals, current supply for microphones and keyswitch, and connection of microphone generated interphone signals to the COM AUDIO output.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of packaging and packing (see Section 5).

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c. When a first article is required (see 4.2).

d. When special tests are required (see 4.4).

6.3 First article. When a first article inspection is required, the C-2646 will be tested and should be a first article sample. The first article should consist of two C-2646s. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles samples. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (keyword) listing.

Active Interphone
AN/AIC-14
C-2646 Set
Control panel
Intercommunications

Preparing Activity:
Navy - AS
(Project 5831-N077)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-85844A (AS)	2. DOCUMENT DATE (YYMMDD) 14 NOVEMBER 1990

3. DOCUMENT TITLE INTERCOMMUNICATIONS SET, AN/AIC-14A ACTIVE INTERPHONE CONTROL PANEL C-2646/AIC-14 GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER NAVAL AIR ENGINEERING CENTER SYSTEMS ENGINEERING & STANDARDIZATION DEPT	b. TELEPHONE (Include Area Code) (1) Commercial (908)323-2326	(2) AUTOVON 624-2326
c. ADDRESS (Include Zip Code) CODE 53 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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